# WAZUH Home Lab - SIEM and File Integrity Monitoring

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**It is encouraged to use the YouTube lab video as a reference when using this guide -** https://youtu.be/QT81wcuoRFY

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# 1. Overview

Wazuh is a free, open-source security platform that offers:

- Log analysis
- File integrity monitoring
- Intrusion detection
- Vulnerability detection
- Real-time alerting

**This guide demonstrates how to build a basic Wazuh setup for learning and experimentation. It is encouraged to use the YouTube lab video as a reference when performing these steps below -** https://youtu.be/QT81wcuoRFY

# 2. Lab Architecture

| Component | Host | Role |
|---|---|---|
| Wazuh Manager | Ubuntu (VirtualBox) | Collects, analyzes, and stores data from agents |
| Wazuh Agent | Windows (host machine) | Sends logs and system events to the Wazuh manager |

**Network Configuration:**

Use **Bridged Adapter** in VirtualBox to place the Ubuntu server on the same network as the host. This allows access between host and guest.

# 3. Prerequisites

- VirtualBox installed
- Ubuntu Server 20.04+ installed in VirtualBox (bridged networking)
- Internet access on Ubuntu VM
- Administrative access on the Windows host
- Optional: basic knowledge of Linux and system administration

# 4. Installing the Wazuh Manager (Ubuntu)

Run the following steps on your Ubuntu VirtualBox server.

## 4.1 Add Wazuh GPG Key

Via a terminal window

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o /usr/share/keyrings/wazuh-archive-keyring.gpg
```

This adds the GPG key to verify Wazuh packages.

### 4.2 Download and Execute Wazuh Installation Script

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash
./wazuh-install.sh -a -i
```

- `-a`: Installs all components (manager, indexer.)
- `-i`: Runs in interactive mode

The script installs all required services and configures them automatically.

# 5. Accessing the Wazuh Dashboard

After installation:

Check your Ubuntu VM's IP address:

```
ifconfig
```

On your Ubuntu server, open a browser and go to:

```
https://<ubuntu-vm-ip>
```

1. Accept any browser security warning due to the self-signed certificate.
2. Log in using the credentials displayed at the end of the installation script.

# 6. Installing the Wazuh Agent (Windows Host)

1. Download the latest Wazuh agent MSI installer from the official documentation:

   [Wazuh Agent for Windows]

2. Install the MSI package on your Windows system using the default settings.

# 7. Registering the Agent with the Manager

### 7.1 Generate Agent Key on Ubuntu Manager

Run the agent management utility:

```
sudo /var/ossec/bin/manage_agents
```

- Select `A` to add an agent.
- Assign a name (e.g., `WindowsHost`).
- Leave IP address blank unless static assignment is needed.
- After creation, select `E` to extract the key.
- Copy the key output.

### 7.2 Apply Key in the Windows Agent

1. Open **Wazuh Agent Manager GUI** from the Start Menu.
2. Paste the copied key into the appropriate field.
3. Save and apply the key.
4. Add the manager's IP address (IP address of your Ubuntu manager).
5. Restart the agent service.

You can then go to the WAZUH dashboard and see the agent onboarded.

# 8. File Integrity Monitoring on Windows

Wazuh supports real-time monitoring of file and folder changes using **Syscheck**.

### 8.1 Edit Agent Configuration

Open the following configuration file:

```
C:\Program Files (x86)\ossec-agent\ossec.conf
```

Add the following entry inside the `Directory` block:

```
<directories realtime="yes">C:\Users\abc\Test</directories>
```

This monitors the specified folder in real-time.

### 8.2 Restart the Agent

After saving the changes, restart the Wazuh agent service to apply the configuration.

# 9. Verifying Setup

1. Open the Wazuh Dashboard in your browser.
2. Navigate to **Agents** → ensure the Windows agent is listed and status is **Active**.
3. Go to the **Integrity Monitoring** section.
4. Perform actions (create/modify/delete files) in the monitored folder.
5. Confirm that alerts appear in the dashboard.